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Analysis of the Multicomponent Magnetization of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada

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A paleomagnetic study on 14 red limestone sites of the Helikian Little Dal Group, 'basinal sequence' (Mackenzie Mountains, Northwest Territories, Canada) resolved five magnetizations using thermal and alternating field treatment with vector analysis. One magnetization (C) with low unblocking temperatures (<350°C) is probably a goethite weathering component of Cretaceous age. It gives a direction at 291°, +75° (12 sites, $k = 73$, $\alpha_{95} = 5°$) and a pole at 60°N, 170°E. Two other closely related magnetizations (A, detrital?) probably carried by magnetite (A_M) and hematite (A_H) respectively, yield directions of 265°,−29° ($k = 87$, $\alpha_{95} = 4°$) and 264°,−26° ($k = 53$, $\alpha_{95} = 6°$) with a combined pole at 16°S,141°E (14 sites, $K = 141$, $A_{95} = 3°$). The two remaining magnetizations (B, red hematite pigment?)–a normal (B_N) and a reverse (B_R) component recognized in most specimens–have a combined direction of 273°,−09° ($k = 17$, $\alpha_{95} = 10°$) and a pole at 3°S,138°E (13 sites, $K = 26$, $A_{95} = 8°$). C and A are prefolding (before Paleocene or pre-late Cretaceous) with A suggested to be primary. A and B lie close to a recently proposed polar track for the late Helikian and Hadrynian and evidence suggests a magnetization age for A of 900 to 950 Ma.

INTRODUCTION

This paper presents results from the first of a series of paleomagnetic studies on the Proterozoic rock succession in the Mackenzie Mountains (Figure 1). Through these studies it is hoped to provide evidence both for the age of these rocks and for their possible relationship with other Proterozoic rocks of the North American craton–especially with possible lithogical equivalents in the Cordillera, and in the Canadian Shield north of Great Bear Lake.

The Proterozoic succession is shown in Figure 2. The lower part of the sequence (the lower part of the Mackenzie Mountains Supergroup) comprising the unnamed H1 unit to the Little Dal Group, and including the basic intrusions which cut all these units, is thought to be largely Helikian in age through tentative correlation of lithologies with the Belt-Purcell Supergroup of southern British Columbia and Alberta and northwestern United States [*Gabrielse,* 1967, 1972]. An unconformity above the Little Dal Group [*Eisbacher,* 1978] supposedly marks the beginning of the Hadrynian succession, which in part includes the remaining rock units of the Mackenzie Mountains Supergroup–the Redstone River and Coppercap Formations. The Rapitan Group, separated from the Coppercap Formation by a profound unconformity [*Gabrielse,* 1967], is thought to be equivalent to the Windermere Supergroup of southwestern British Columbia. Sedimentation of the Belt-Purcell and Windermere rocks probably took place within the broad limits of about 1600 to 850 Ma and 850 to 570 Ma respectively [*Gabrielse,* 1972; *Hofmann and Aitken,* 1979]. Additional lithological correlations have also been made with rocks north of Great Bear Lake: those of the Brock Inlier, the Coppermine area, and Victoria and Banks Islands [see *Young,* 1977]; but these stratigraphic successions like those of the Mackenzie Mountains have not been precisely dated.

Attempts have been made recently to make biostratigraphic correlations between parts of the Proterozoic succession of the Mackenzie Mountains and dated successions in other parts of the world. Correlation of algal stromatolite assemblages with those in the Soviet Union suggests an age of about 1350 Ma for map-unit H1, and an age of about 950 Ma for the 'upper carbonate subunit' of the Little Dal Group [*Aitken et al.,* 1978]. *Hofmann and Aitken* [1979] suggested an age of 1100 to 800 Ma for the 'basinal sequence' of the Little Dal, based on plant microfossils and macrofossils. The older limit suggested by them was, however, much influenced by the then-known range of *Chuaria,* now known to range back into strata as old as 1600 Ma in China (J. D. Aitken, personal communication, 1979). The only pertinent radiometric dates available are Rb-Sr isochron dates of 766 ± 24 Ma and 769 ± 27 Ma ($\lambda = 1.42 \times 10^{-11}$ yr^{-1}) from two gabbroic sills in the Tsezotene Formation which are probably related to intrusions that cut the 'basinal sequence' and at least as high as the overlying 'grainstone subunit' (R. L. Armstrong et al., manuscript in preparation 1980; J. D. Aitken, personal communication, 1979). *Young et al.* [1979] assign the Little Dal Group to a supra-regional sequence deposited between ~1.2 and ~0.8 Ga ago.

All the major units of the Helikian succession save H1 were sampled, as well as possible correlatives in the Brock Inlier (Figure 2); but only results from the Little Dal Group are presented here. Major folding occurred in the Helikian succession of the Mackenzie Mountains in Paleocene or pre-late Cretaceous times. Displacements of the Little Dal Group at the Mackenzie Mountain Front relative to the Canadian Shield are estimated to have been less than 10 km in total [*Aitken et al.,* 1980]. The Little Dal Group is composed of six formations [*Aitken,* 1977; *Aitken et al.,* 1978]. Only the two oldest units, the so-called 'mudcracked subunit' and the conformably overlying 'basinal sequence' were sampled; both from each limb of a general synclinal structure located as shown in Figures 1 and 3. The samples of the present study were collected from one informal member of the 'basinal sequence' of the Little Dal, namely, the so-called 'Dead End shale' [*Aitken,* 1977; *Aitken et al.,* 1978]. Nine sites were collected from one limb of the fold (Rouge Mountain River locality) and five from the other (Dodo Canyon locality). The 'Dead End shale' consists of 'brick red to deep brownish red, calcareous laminated shale with abundant ellipsoidal nodules and nodular thin beds of red to pinkish-gray, microcrystalline limestone' [*Aitken and Cook,* 1974]. Samples were obtained

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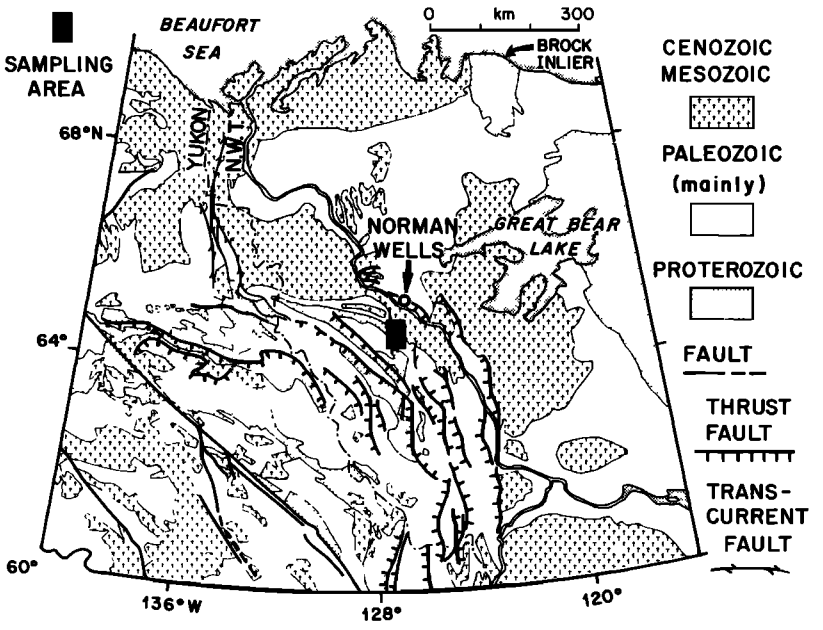


Fig. 1. Location map showing the area of sampling (solid rectangle) in the Mackenzie Mountains of the Canadian Arctic.

from the nodules, each site from a separate layer ranging from one to ten metres apart within a total stratigraphic thickness of 40 to 50 m. Five sandstone and one gray limestone site obtained from the lowest formation, and antoher site from the top of the Rouge Mountain River 'basinal sequence' which displayed divergent magnetic behavior, will be discussed in a subsequent paper.

METHODS

Block samples (nodules) were collected in the field and drilled in the laboratory; two specimens being generally obtained from each sample. Experimental studies involved treatments by alternating fields (AF)–usually up to 20 mT–and (or) heating carried out in quasi-zero (<2nT) residual fields [*Roy et al.*, 1969]. Most specimens have been demagnetized using the combined treatments; however, it was realized late in the study that this was a less effective method than straight thermal work for revealing the magnetizations present. Specimens were thermally demagnetized with a Schonstedt TSD-1 and AF treated on the apparatus described by *Roy et al.* [1973]. They were usually measured with Schonstedt spinner magnetometers (DSM-1), but occasionally with a Geofyzika JR4 spinner. Since all specimens from each site are from a single layer, and because of the difficulties in analyzing the complex NRM (natural remanent magnetization), specimen rather than sample data are used as the basic within-site data.

The complex magnetization was analyzed by vector subtraction (VS) techniques, and where possible, verified through the use of vector diagrams similar to those described by *Zijderveld* [1967] and by *Roy and Park* [1974]. Both types of vec-

TABLE OF FORMATIONS

MACKENZIE MTNS.		BROCK INLIER	
RAPITAN GROUP	●		
COPPERCAP FM.			
REDSTONE RIVER FM.			
BASIC INTRUSIONS	▲		
LITTLE DAL GROUP	●	P4	
KATHERINE GROUP	▲	P3	▲
TSEZOTENE FM.	▲		
H1 UNIT		P2	▲

▲ SAMPLED
● RESULTS REPORTED
UNCONFORMITY

Fig. 2. Rock units of the Mackenzie Mountains area and possible equivalents in the Brock Inlier [*Gabrielse*, 1972; *Aitken et al.*, 1973; *Young*, 1977].

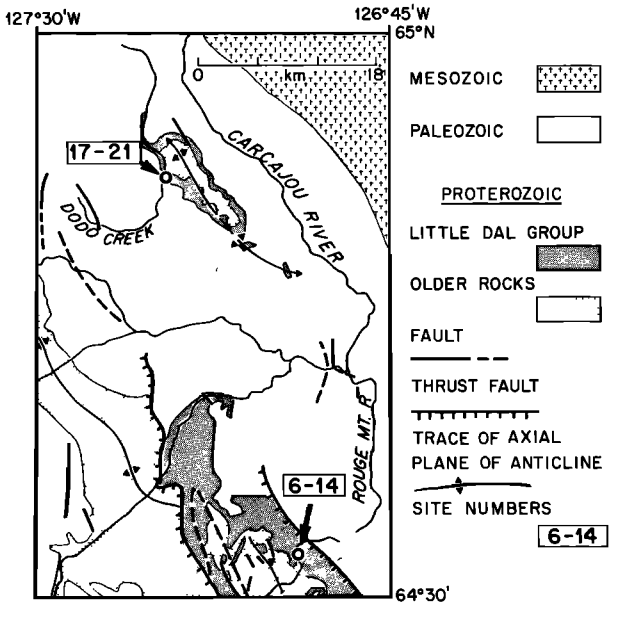


Fig. 3. Sampling localities and associated sites.

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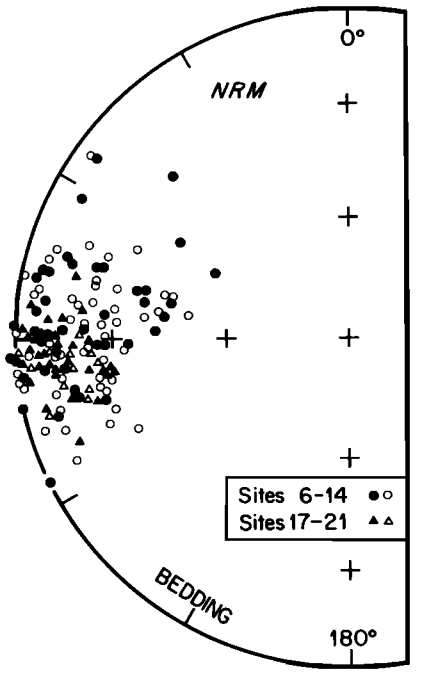


Fig. 4. All specimen natural remanent magnetizations (NRMs) plotted with respect to bedding on an equal area stereographic projection. Open (closed) circles are plotted on upper (lower) hemisphere. In general, open or closed symbols have this meaning in succeeding figures.

tor diagram display vectors on the XY plane, but the latter (used in this paper) displays the vectors on the vertical ZH rather than the ZX or ZY planes, thus allowing the inclination (like the declination in the XY plane) to be read directly. Vector diagrams are useful because single-component magnetic directions are ideally revealed as straight line segments. Sometimes though, straight lines may represent mixed components which are being removed in the same proportion under treatment. Results obtained using both VS and vector diagrams are almost the same and may be compared in several of the tables.

The use of VS generally relies on the observation that only one magnetic component changes during a given treatment interval. Vector diagrams will provide a ready verification of this fact. The presence of two 'endpoints' or apparent endpoints (AE) on stereograms points to the removal of essentially one magnetization within the cleaning interval between them. Here the term apparent signifies that more than one undemagnetized component remains following the removal of a less-stable component. The removed component (in this study, A_M, see below) is found by vectorially subtracting one AE from the other.

There is at least one case where VS can resolve two magnetizations which are being concurrently removed within a given treatment interval; namely, the instance where the two magnetizations are nearly antiparallel (say within 20°). They may even have nearly complete overlap of unblocking temperature (T_{UB}) or resistive coercive force (rcf) spectra. The technique

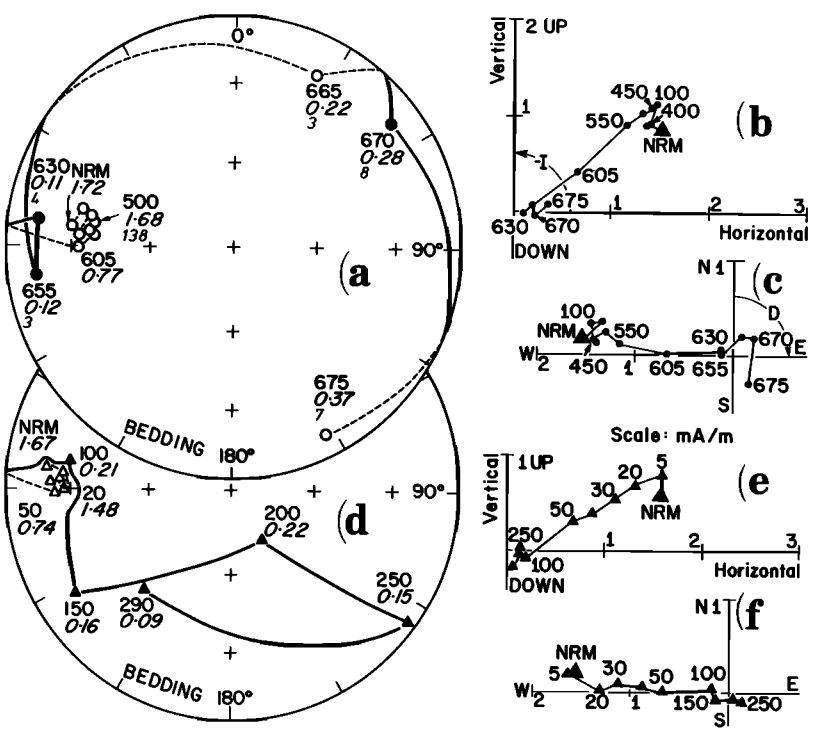


Fig. 5. Comparison of the magnetic direction changes in two specimens from the same sample (site 9), one treated thermally (*a*, *b*, *c*) and the other by alternating fields (*d*, *e*, *f*). The vector diagrams depict the magnetizations in the vertical plane (*b*, *e*) and the horizontal plane (*c*, *f*), where *I* is the inclination and *D* is the declination. The intensities are represented by the vector length from the origin. All vector diagrams have been corrected for bedding. The groups of numbers in (*a*) and (*d*) represent from top to bottom the treatment (°C (*a*) or mT (*d*)), the intensity (mA/m), and an internal precision parameter showing the within-specimen uniformity of direction. In Figure 5–9 circles represent AF data and triangles thermal data.

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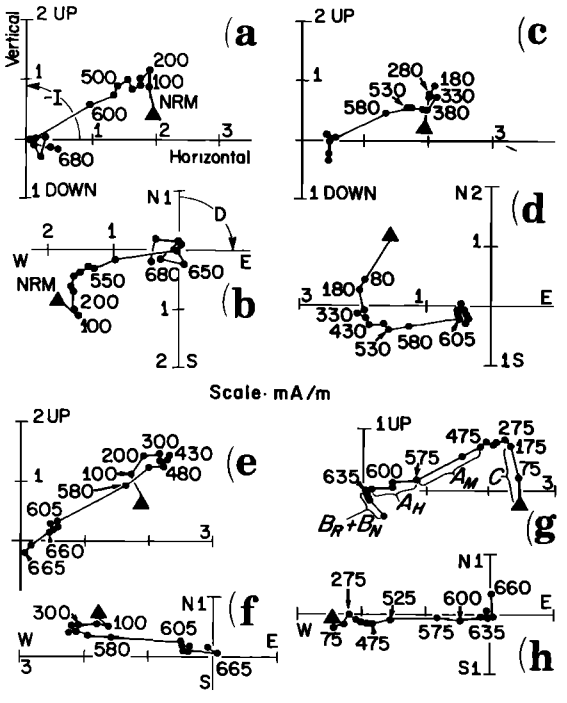


Fig. 6. Vector diagrams illustrating the pure thermal treatment of specimens from sites 7 (*a*, *b*), 8 (*c*, *d*), 10 (*e*, *f*), and 19 (*g*, *h*). Part (*g*) depicts generally the T_{UB} ranges of magnetic components that have been resolved in this study. There is probably an additional, unresolved, component with the T_{UB} range 100 to 500°C (see for instance (*c*) and (*e*)).

for analyzing this case has been fully explained in *Park and Roy* [1979]. It relies on small relative changes between the rates of decay of the components. Because the resultant is small compared to its components, these changes can produce relatively large changes in the direction and intensity of the resultant. As the resultant approaches one or other of the component directions both the intensity and internal precision (related to the homogeneity of the direction of magnetization) of the magnetization increase. The best approximation to the true direction (irrespecitve of sign) is found by subtraction between vectors which have relative maximum or minimum values of intensity or direction on intensity decay curves, stereoplots, or vector diagrams. Any one determination will not yield a true direction unless only one of the components changes, but an average of many VS directions should approach the true direction. The use of vector diagrams will not directly resolve this case except to indicate which vectors are to be subtracted. In this study the directions B_N and B_R (see below) have been found in this manner.

Vector diagrams of this study are often slightly irregular in the 'linear' segments. Several factors may account for this: there may be (1) a slight overlap of components, (2) an effect due to secular variation of the earth's field, or (3) effects due to magnetic noise. Factor (2) is possible in a slowly accumulating and drying (compacting) sediment, and may relate to the present rocks. The rationale behind this proposed process is that larger, and therefore lower T_{UB}, grains would be magnetized earlier in the earth's field, whereas the smaller, higher T_{UB}, grains would be free to respond to the later ambient field until the sediment became dryer [*Verosub*, 1977]. Factor (3) is also a distinct possibility in this study. The specimens had NRM intensities of <3 mA/m and treated values generally <1 mA/m; and precautions had to be taken (such as subtracting the intensity of sample holders) in order to eliminate major noise components. It should also be pointed out that deviations owing to noise are much more evident in vector diagrams than in stereograms which do not show differences in intensity.

Results and Analysis

Specimen NRMs are plotted with respect to bedding in Figure 4. Under thermal and AF treatment a number of different magnetizations were uncovered and their directions resolved using the techniques described above. AF work revealed one and often two AEs: the first from 20 to 50 mT (Figures 5*d*–*f*) and the second, less-accurately defined, in the range 50 to 250 mT (not figured). Thermal demagnetization showed evidence of at least five and possibly six magnetizations altogether (Figures 6 and 7; see Figure 6*g* for the general unblocking temperature ranges of resolvable components). The lowest unblocking temperature (T_{UB}) component (*C*) occurs between the NRM and about 200°C in vector diagrams (e.g., Figure 6), but is not apparent on most stereoplots. The next two resolvable components are illustrated by the presence of two very closely-spaced AEs in the temperature range 500 to 635°C–the lower at ca. 525°C (Figure 7*b*),–and they are clearly separate entities with reference to the intensity decay plot, Figure 7*a*. In most vector plots the two magnetizations which respectively characterize each AE (A_M and A_H) are almost colinear. Another component (*D*) occurring in the range between about 100 and 500°C appears to be intimately mixed with *C* and A_M over part of their T_{UB} ranges and cannot be resolved. Combined AF and thermal treatments revealed all the magnet-

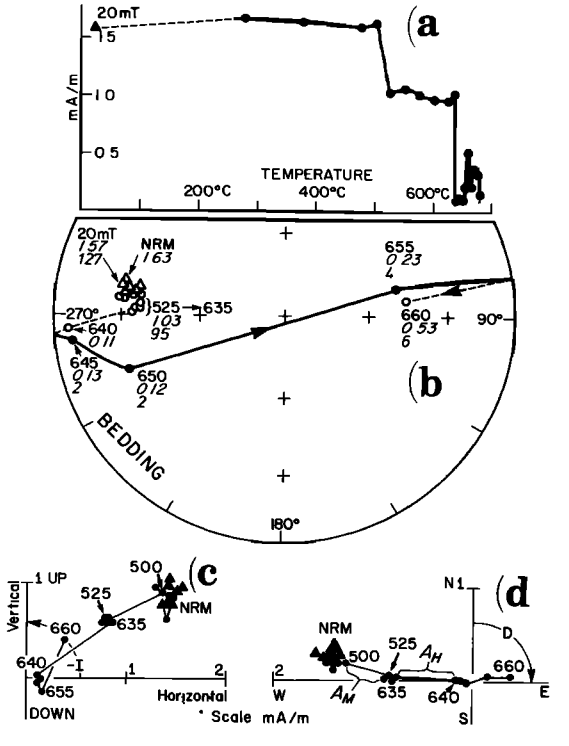


Fig. 7. Ideal example showing the thermal treatment of a specimen from site 9 previously treated in an AF of 20 mT. (*a*) shows the intensity decay curve, (*b*) the stereogram, and (*c*) and (*d*) the vector diagrams. The main T_{UB} ranges of A_M and A_H are indicated. See Figure 5 for explanation of symbols.

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izations that were observed under separate treatments except where specimens were initially treated in high AFs (100 to 250 mT). In this case, high AFs eliminated A_M which characterizes the 525°C AE. This fact combined with the observation that low AFs have little effect upon magnetic components with T_{UB}'s >600°C revealed the possibility of determining A_M by both AF and thermal methods. The recovery of this common magnetization from both AF and thermal data is explained below.

Further thermal cleaning of specimens above about 635°C generally revealed a magnetic signature which indicated the existence of at least two additional magnetizations (B_R and B_N). Figure 8 provides a good illustration. Here, the magnetic vector decreases in intensity to 0.26 mA/m at 670°C before radically changing direction to the east at 675°C with an increase in intensity of 0.77 mA/m. Thereafter the intensity passes through another minimum before returning to the westerly direction with increased intensity. There is often an attendent direct increase or decrease with intensity of the homogeneity of the magnetization in the specimen (e.g., see Figures 5, 7 and 8). The presence of this pattern of magnetic behavior is a good indication of the presence of sub-antiparallel magnetizations having T_{UB} or rcf spectra which substantially overlap. These magnetizations have been analyzed according to the technique outlined above. This pattern of behavior is not always observed, however, even when a vector changes progressively to an antiparallel direction, because (1) the number of treatment steps may not be enough to locate the areas where true maximum and minimum intensity directions occur, and (2) there may be an additional superimposed magnetization (e.g., A_H). An example illustrating the resolution of all these components (C, A_M, A_H, B_R, B_N) by VS is given in Figure 9 and Table 1.

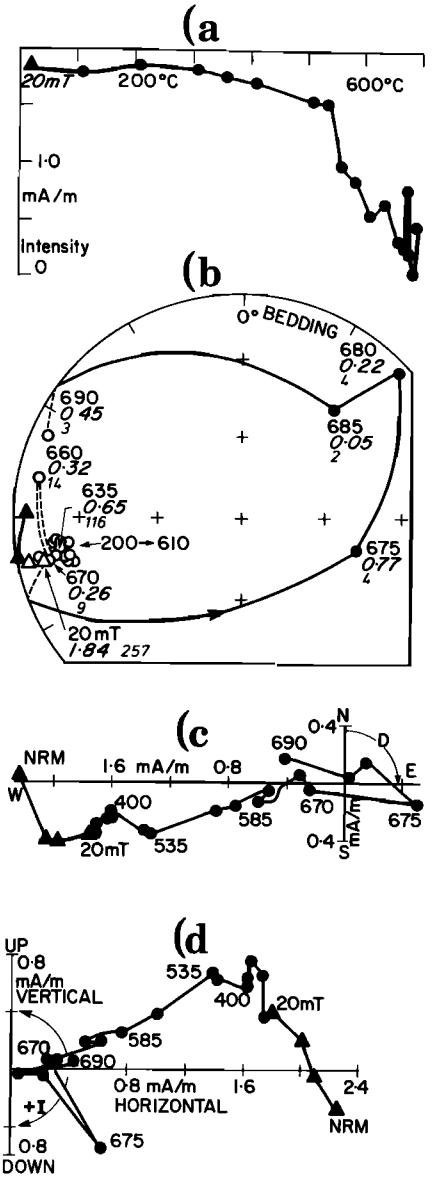


Fig. 8. The combined AF and thermal treatment of a specimen from site 20 illustrating the migration of the direction to B_N (east) and then to B_R (west). (*a*) depicts the intensity decay curve which in contrast to that of Figure 7 shows no clear separation of A_H and B components. (*b*) Note how the intensity decreases to a low value (0.26 mA/m) at 670°C, increases (0.77) at 675°C in the general B_N direction, and again decreases (0.05 at 685°C) before increasing again (0.45) at 690°C in the general B_R direction. (*c*) and (*d*) show the vector diagrams which more readily illustrate the presence of the three hematite components A_H, B_N, and B_R. A_M is mainly eliminated from 535 to 586°C, A_H from 585 to 670°C, and B_R and B_N from 670 to 690°C. See Figure 5 for explanation of symbols.

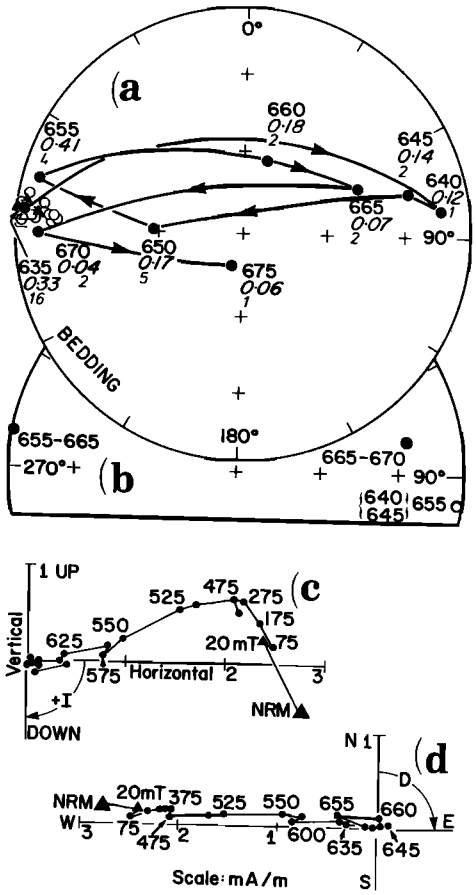


Fig. 9. Thermal treatment of a specimen from site 18 previously AF treated in 20 mT, illustrating the typical behavior encountered in rocks containing normal and reverse components with similar T_{UB} ranges. The NRM (2.82 mA/m) is represented by the easternmost solid triangle. (*b*) shows the VS directions of B_N (east) and B_R (west) (see Table 1). The two B_N directions would normally be averaged. Directions and intensities of the 640 and 645°C vectors have been meaned before VS of the 655°C vector. See Figure 5 for explanation of symbols.

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TABLE 1. Example of Vector Subtraction to Determine Magnetizations

	First Vector			Second Vector			Resultant Vector (R)		
	Trt, °C	M	$D°, I°$	Trt, °C	M	$D°, I°$	M	$D°, I°$	Entry
A_H	575	0.77	276, −03	635	0.12	082, +14	0.88	274, −05	1
B_N	640,645	0.13	079, +21	655	0.41	283, +09	0.52	098, −02	2
B_N	665	0.07	068, +45	670	0.04	269, +11	0.10	078, +25	3
B_R	655	0.41	283, +09	665	0.07	068, +45	0.45	279, +02	4
A_M	525	2.10*	273, −20	575	0.77	276, −03	1.39	271, −29	5

The Resultant Vector magnetization **R** (A_H, B_N, B_R, or A_M) with respect to bedding is obtained by subtracting the Second Vector from the First Vector. The vector directions, their associated treatments (Trt), and the **R** values are shown in Figure 9. Entries 2 and 3 would be averaged together. M, $D°$, and $I°$, are respectively the intensity (mA/m), declination, and inclination of the vector. The temperatures indicated are approximate.

*Correction made to intensity for decrease to magnetite component caused by AF of 20 mT.

C Magnetization

The first magnetization revealed (labelled *C*) is best illustrated in the vector diagrams of straight thermally-cleaned specimens (Figure 6), which reveal the removal of a low T_{UB} component between the ambient temperature and about 200°C, with a resulting slight increase in total intensity. Some irregularities in the vector diagrams in the temperature range between 100 to 500°C, indicate the possible presence of a steeply reverse magnetization (*D*) (Figure 6, see also Figure 8*d*). These features are found quite generally, but have not been analyzed owing to the apparent mixture of components *C*, *D*, and A_M in this temperature range.

C has been found directly from Zijderveld diagrams, but is possibly slightly contaminated by the *D* component. The site results given in Table 2 are plotted both with respect to horizontal (Figure 10*a*) and to bedding (Figure 10*b*). An average is given in Table 7.

Several lines of evidence indicate that this magnetization is carried by goethite or a related mineral. First, the low T_{UB} component, according to the vector diagrams, apparently does not extend beyond about 200°C, which is within the goethite T_{UB} range. Second, on some stereograms depicting thermal cleaning following high AF treatment, successive thermal directions appear to cluster about the last AF point (250 or 290 mT) until between 300 and 400°C is reached. This is the T_{UB} range within which goethite inverts to hematite. Other experiments, involving the progressive acquisition of partial thermoremanent magnetization during cooling in a small bias field (0.053 mT), also confirm these temperatures by revealing a magnetization which peaks at about 200°C and apparently disappears between 350 and 365°C. Third, this magnetization is largely unaffected by low AFs (even fields up to 290 mT) in accord with the fact that goethite can have quite high coercive forces (e.g., 100 to 400 mT) [*Vlasov et al.*, 1967; *Bagin et al.*, 1976]. Figure 11*a* shows that low fields of 20 or 30 mT mainly eliminate part of the *A* magnetization (see below) with unblocking temperatures between 500 and 600°C (Rouge Mountain River locality) rather than the *C* magnetization. And finally, thermomagnetic work on powders indicates that an increase in magnetization occurs within the main range of hematite T_{UB}'s, perhaps due to the acquisition of magnetization by hematite formed from goethite.

TABLE 2. *C* Direction With Respect to Bedding

Site	N	$D°, I°$	R	$\alpha_{95}°$
	Rouge Mountain River Locality			
7	2	305, +63	1.99	...
8	...	... ...	...	...
9	...	... ...	...	...
10	1	266, +74	...	...
11	2	300, +64	1.83	...
12	1	335, +76	...	...
13	2	269, +76	1.93	...
14	5	329, +74	4.89	13
6	3	293, +71	2.73	49
	7	*300, +72*	*6.92*	*7*
	Dodo Canyon Locality			
21	3	300, +69	2.93	24
20	4	206, +86	3.92	16
19	4	258, +76	3.91	17
18	2	281, +74	2.00	5
17	4	266, +75	3.78	26
	5	*275, +77*	*4.95*	*8*

N is the number of specimens, R, the vector resultant; and $\alpha_{95}°$, the half-angle of cone of 95% confidence about the direction. Other symbols are explained in Table 1. Averages of localities are given in italics.

A and B Magnetizations

The next magnetization revealed, with T_{UB}'s mainly between 500 and 600°C, is clearly evident in Figure 7 by the AE (apparent endpoint) at ca. 525°C. Closely associated with it is another magnetization with T_{UB}'s usually <640°C (but some-

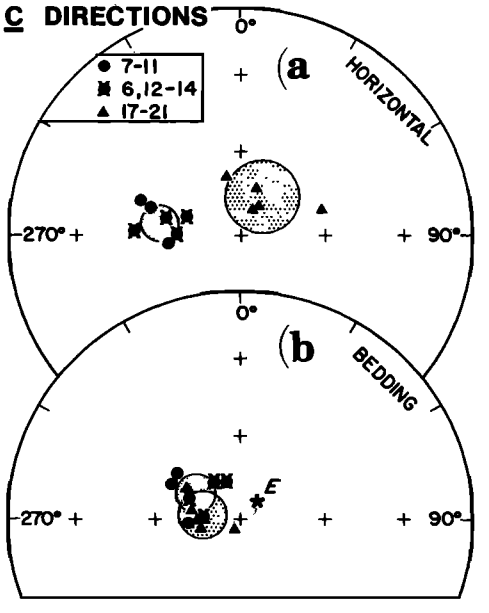


Fig. 10. Fold test applied to the *C* site directions. The ellipses are the α_{95} errors about the mean directions.

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TABLE 3. Comparison of Magnetite Site Vectors (A_M) Determined by Three Different VS Methods and from Vector Diagrams

Entry	Site	Method 1: $A_M = A_M' - A_H'$					Method 2: $A_M = A_M' - A_H - B_R - B_N$					Method 3: $A_M = A_M' - A_H - B_N$					Method 4: Zijderveld Diagrams			
		N	*M*	*D°, I°*	*R*	α_{95}°	*N*	*M*	*D°, I°*	*R*	α_{95}°	*N*	*M*	*D°, I°*	*R*	α_{95}°	*N*	*D°, I°*	*R*	α_{95}°
									Rouge Mountain River Locality											
	7	5	1.03	265, −30	4.97	7		1.13	255, −36				1.47	260, −22			9	264, −34	8.89	6
	8	8	1.26	266, −36	7.72	11		1.16	271, −26				1.56	265, −26			10	265, −35	9.64	10
	9	7	0.83	275, −37	6.93	6		(0.87	265, −40)				...	...			10	272, −35	9.91	5
	10	8	1.86	274, −35	7.94	5		(1.90	270, −38)				...	...			8	272, −37	7.94	5
	11	8	0.88	266, −34	7.96	4		0.78	267, −33				0.96	273, −29			9	263, −33	8.88	6
1		*5*	*1.17*	*269, −34*	*4.99*	*5*	*5*	*1.17*	*266, −35*	*4.97*	*7*						*5*	*267, −35*	*4.99*	*4*
2							*3*	*1.02*	*265, −32*	*2.98*	*13*	*3*	*1.33*	*266, −26*	*2.99*	*10*				
	12	8	0.96	266, −32	7.95	5		0.68	278, −28				1.34	267, −19			8	263, −31	7.95	5
	13	5	1.52	260, −20	4.96	8		...	... ...				...	...			10	264, −19	9.74	8
	14	10	1.19	265, −14	9.92	5		1.00	258, +01				2.16	257, −06			11	262, −17	10.87	5
	6	4	1.69	255, −34	3.97	9		2.00	228, −18				3.44	262, −20			10	258, −33	9.87	6
3		*4*	*1.34*	*262, −25*	*3.95*	*12*											*4*	*262, −25*	*3.97*	*10*
4		*3*	*1.28*	*262, −27*	*2.95*	*19*	*3*	*1.23*	*254, −16*			*3*	*2.31*	*262, −15*	*2.97*	*14*				
5		**9**	**1.25**	**266, −30**	**8.89**	**6**	*6*	*1.13*	*259, −24*	*5.67*	*18*	*6*	*1.82*	*264, −20*	*5.93*	*8*	*9*	*265, −31*	*8.92*	*5*
6		*8*	*1.21*	*266, −32*	*7.91*	*6*	*8*	*1.19*	*261, −28*	*7.61*	*18*									
									Dodo Canyon Locality											
	21	8	1.99	261, −36	7.95	5		1.89	270, −29				2.75	262, −29			11	260, −36	10.96	3
	20	8	1.14	260, −25	7.96	4		0.65	246, +10				1.45	263, −21			9	261, −25	8.95	4
	19	7	1.02	261, −24	6.96	5		1.52	260, −21				2.59	260, −29			7	263, −27	6.90	8
	18	7	1.08	271, −26	6.91	8		1.46	275, −28				1.77	277, −21			8	268, −25	7.93	6
	17	7	1.73	264, −21	6.90	8		1.53	274, −14				2.29	271, −16			8	266, −24	7.82	9
7		**5**	**1.39**	**263, −27**	**4.97**	**7**	*5*	*1.41*	*265, −17*	*4.76*	*19*	*5*	*2.17*	*267, −23*	*4.95*	*8*	*5*	*264, −27*	*4.98*	*5*
8		**14**	**1.30**	**265, −29**	**13.85**	**4**	*11*	*1.25*	*262, −21*	*10.40*	*12*	*11*	*1.98*	*265, −22*	*10.88*	*5*	*14*	*264, −29*	*13.89*	*4*
9		*13*	*1.28*	*265, −30*	*12.87*	*4*	*13*	*1.27*	*263, −24*	*12.31*	*17*									

Vectors used for calculating A_M: A_M' is A_M prior to correction; A_H', A_H prior to correction; A_H, the hematite A vector; B_R and B_N, the reverse and normal B vectors. Other symbols are explained in Tables 1 and 2. See Tables 5 and 6 for the calculated values of B_R, B_N, and A_H used in Methods 2 and 3. See text for a discussion of the methods. The preferred data calculated by VS Method 1 may be compared to that calculated from vector diagrams (Method 4). Averages of sites are in italics or boldface; those in boldface are the preferred averages. Comparisons between Methods 1 and 2 can be made using entries 1, 4, 6, 7, and 9; and between Methods 2 and 3 by using entries 2, 4, 5, 7, and 8. Parentheses indicate estimated values. Directions with respect to bedding.

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TABLE 4. A_M Determined Using Alternating Field Treatment

Site	Vector Subtraction N	Vector Subtraction M	Vector Subtraction D°, I°	Vector Diagrams N	Vector Diagrams D°, I°
		Rouge Mountain River Locality			
7	1	1.03	242, −24	...	...
8	2	0.92	256, −26	...	...
9	1	1.38	268, −35	1	272, −36
10	2	1.59	269, −27	...	...
11	3	0.78	256, −26	...	...
12	...	...	...	...	...
13	3	1.76	265, −25	1	270, −22
14	2	1.01	263, −06	...	...
6	2	1.59	259, −28	2	261, −27
	8	*1.26*	*260, −25*		
		(7.86, 51, 8)			
		Dodo Canyon Locality			
21	...	...	...	2	262, −34
20	1	1.09	266, −30	...	...
19	1	1.03	258, −17	...	...
18	2	1.27	272, −31	...	...
17	4	1.57	264, −27	1	261, −24
	4	*1.24*	*265, −26*		
		(3.97, 97, 9)			
	12	**1.25**	**261, −25**		
		(11.82, 63, 6)			

N is the number of specimens or sites (site in italics). The VS values were obtained by subtracting the apparent endpoint at 50 to 200 mT from that at 20 mT. Values obtained from vector diagrams are given for comparison. Locality averages are in italics and grand average in boldface. *R*, the vector resultant, *k*, the precision, and α_{95}, the half-angle of cone of 95% confidence about the direction are given in parentheses. Compare with A_M determined by VS of thermal vectors (Tables 3 and 7).

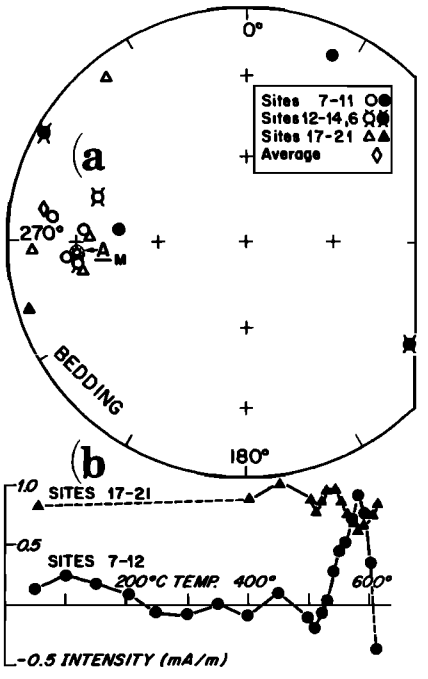


Fig. 11. The magnetization eliminated by AFs of 20 to 30 mT. (*a*) The average direction of magnetization eliminated at 525°C lies close to A_M. These directions were obtained by averaging the vectors of those specimens (4 to 8 per site) each treated in AFs and 525°C, and subtracting this value from the average vector of those specimens treated at 525°C only (3 per site). (*b*) T_{UB} spectrum of that magnetization eliminated by AF. This is the average difference curve between companion specimens (from one sample per site): one treated by an AF prior to thermal demagnetization. The curves of the purely thermal specimens have been normalized to the NRM intensity prior to subtraction. Much of the magnetization removed by low AFs from specimens of the Rouge Mountain River locality (sites 7–12) has T_{UB}'s between 500 and 600°C characteristic of magnetite.

times as high as 675°C) which does not change direction to any extent until at least after the 635°C step (Figure 7). The distinction between these two components on diagramatic representations is not always clear (e.g., see Figure 8) but sometimes (Figure 7); both are nominated *A*.

Thermomagnetic curves reveal a Curie point at about 580°C, which is probably due to magnetite, and likely corresponds to the less-stable *A* component (labelled A_M). The other component (labelled A_H) would seem to be 'hematite',–perhaps titanohematite, considering its usually comparatively low T_{UB}'s,–though no Curie point is indicated on the thermo-

TABLE 5. Hematite Site Vectors (A_H), and Combined Hematite and Magnetite Site Directions With Respect to Bedding

Site	N	A_H From Vector Subtraction M	D°, I°	R	α_{95}°	A_H From Vector Diagrams N	D°, I°	R	α_{95}°	Combined A_H and A_M: A D°, I°	Pole (°S, °E)
						Rouge Mountain River Locality (64.5°N,126.9°W)					
7	3	0.65	272, −30	2.90	29	3	271, −32	2.90	28	268, −31	16, 137
8	7	0.70	265, −24	6.57	17	6	263, −38	5.73	16	265, −37	21, 139
9	8	0.70	269, −38	7.96	4	8	271, −37	7.97	3	273, −37	17, 131
10	7	0.54	276, −29	6.67	15	8	276, −30	7.95	5	275, −32	14, 131
11	8	0.73	260, −23	7.89	7	8	260, −22	7.83	9	263, −28	16, 143
12	8	1.34	263, −21	7.84	9	5	261, −21	4.97	7	264, −26	15, 143
13	7	1.19	264, −12	6.84	10	8	260, −20	7.85	8	260, −20	14, 148
14	6	0.50	271, −04	5.44	24	7	267, −03	6.77	12	266, −09	06, 145
6	2	0.47	257, −34	1.74	...	1	247, −32	...	...	255, −34	23, 148
						Dodo Canyon Locality (64.9°N,127.2°W)					
21	6	0.74	260, −36	5.91	9	8	263, −36	7.86	8	262, −36	21, 141
20	7	0.43	266, −18	6.90	8	7	260, −26	6.86	9	260, −26	16, 146
19	7	0.53	261, −18	6.78	12	7	264, −25	6.83	10	262, −25	15, 144
18	5	0.94	266, −17	4.90	12	8	267, −17	7.79	10	269, −21	11, 139
17	6	0.78	255, −21	5.76	15	7	263, −27	6.94	6	264, −24	14, 143

See Tables 2 and 3 for explanation of symbols. A_H was calculated by subtracting from A_H' the adjacent minimum intensity vector, the resultant of B_R and B_N following the apparent demagnetization of A_H. The *A* values are the means of A_M (Method 1, Table 4) and A_H (from vector diagrams) combined at the site level, except for site 6 where only the VS values have been combined.

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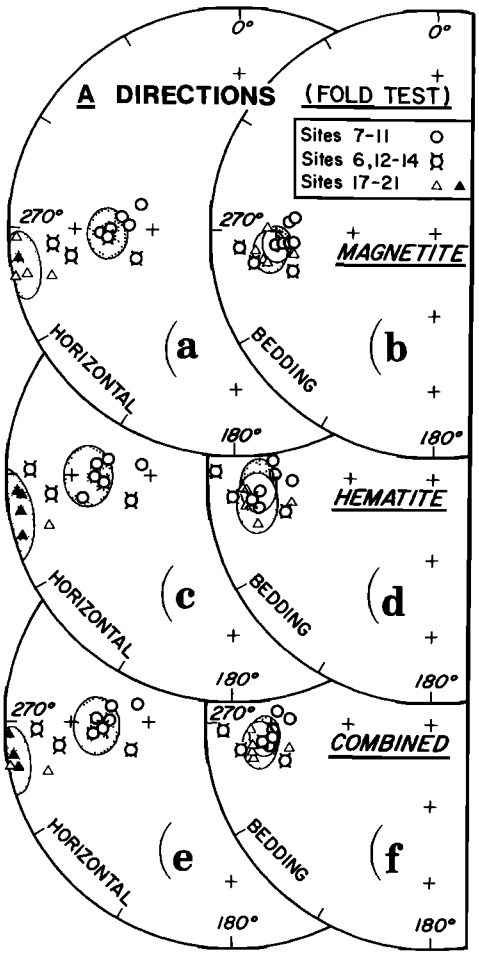


Fig. 12. Fold test on the A site directions of magnetite (a), hematite (b), and magnetite and hematite combined (by site) (c). The ellipses are the α_{95} errors about the mean directions. These directions are from VS only.

magnetic curves. A_M appears to be definitely affected by low AFs (20 or 30 mT) (Figure 11b),—particularly in the Rouge Mountain River samples,–and to be completely removed in high AFs (100 to 250 mT) leaving A_H intact.

Upon further thermal demagnetization, usually above 635°C, the magnetic behavior indicates the presence of sub-antiparallel magnetizations as mentioned earlier. Directions migrate to the east (Figures 5, 7, and 9) and then back to the west (Figures 8 and 9), or exhibit erratic behaviour. Intensities in this temperature range (ca. 635 to 690°C) are generally weak (<0.5 mA/m). Only one clear example of a similar migration under AF cleaning was documented (Figure 5b). The normal (B_N) and reverse (B_R) components exhibiting this general behavior are nominated the B magnetization. The B components apparently have similar T_{UB} ranges (ca. 635 to 690°C) and like A_H are probably carried by hematite. It is thought, partly on the basis of the distinctly separate behavior patterns in the hematite range of T_{UB}'s (Figure 7), that A_H and B are indeed different magnetizations and can in most cases be largely separated from one another. This matter is discussed in the next section.

A_M and A_H, respectively indicated by A'_M–the AE vector between 500 and 600°C,–and A'_H–the AE vector revealed after the elimination of A'_M,–have been obtained by VS and verified by data from Zijderveld diagrams. A specific specimen example is given in Table 1. A_M was found by subtracting A'_H from A'_M (Method 1, Table 3), and the direction agrees almost exactly with that obtained from the vector diagrams (Method 4, Table 3).

A_M was also revealed under straight AF treatment by VS of the AE at about 100 mT (50 to 200 mT) from that at 20 mT (Table 4), but only the more reliable thermal data has been used for the final averages (Table 7). A comparison of AF with two-stage results may be made by comparing Tables 4 and 3.

A_H was found by subtracting from A_H' the adjacent minimum intensity vector (e.g., Figure 7: 635°C–640°C; Table 5), the resultant of $\mathbf{B}_R$ and $\mathbf{B}_N$ following the apparent elimination of A_H. A_M, A_H, and their combined averages are plotted in Figure 12, both with respect to horizontal (a, c, e) and to bedding (b, d, f), and site averages and grand averages given in Tables 3, 5, and 7.

B directions were obtained by the general VS method described in *Park and Roy* [1979]. The average B_N and B_R site directions (Table 6) are plotted in Figure 13. Their grand averages corrected for bedding are almost 180° apart (271°,

TABLE 6. B Site Vectors With Respect to Bedding

	$\mathbf{B}_M$					$\mathbf{B}_R$					B		
Site	N	M	$D°, I°$	R	$\alpha_{95}°$	N	M	$D°, I°$	R	$\alpha_{95}°$	M	$D°, I°$	Pole (°S, °E)
Rouge Mountain River Locality (64.5°N,126.9°W)													
7	5	0.46	071, −03	4.69	22	2	0.47	271, +13	1.97	44	0.46	261, +08	0, 153
8	3	0.33	091, −10	2.64	58	1	0.42	247, −24	...	...	0.38	260, −07	08, 151
9	...	...	...	...	...	3	0.43	283, +06	2.58	65	0.43	283, +06	−08, 132
10	...	...	...	...	...	1	0.48	285, +13	...	...	0.48	285, +13	−12, 132
11	2	0.18	121, +18	1.96	52	3	0.20	292, −10	2.86	34	0.19	296, −14	−05, 116
12	1	0.49	106, −14	...	...	4	0.69	258, −09	3.80	25	0.59	272, +02	−02, 142
13	...	...	...	...	...	...	...	...	...	...	...	...	...
14	4	0.93	073, 0	3.60	36	3	1.17	257, −12	2.84	37	1.06	255, −06	09, 156
6	3	1.63	097, −01	2.71	51	3	2.05	295, −17	2.92	25	1.84	286, −08	−03, 127
Dodo Canyon Locality (64.9°N, 127.2°W)													
21	4	0.73	083, +17	3.70	30	1	0.90	245, −29	...	...	0.81	255, −23	18, 152
20	3	0.45	093, +12	2.80	42	1	0.98	277, −39	...	...	0.71	275, −26	11, 132
19	2	1.20	081, +38	1.94	64	2	1.11	260, −41	1.96	50	1.15	261, −40	24, 141
18	4	0.54	101, +19	3.71	30	2	0.37	284, +08	1.95	60	0.45	282, −05	−03, 130
17	3	0.50	111, +18	2.82	39	4	0.76	264, −19	3.56	38	0.63	278, −19	06, 132

See Tables 2 and 3 for explanation of symbols. $\mathbf{B}_R$ and $\mathbf{B}_N$ have been calculated using the VS method of *Park and Roy* [1979] as generally indicated in the text. **B** is the simple mean of $\mathbf{B}_N$ and $\mathbf{B}_R$. Each specimen direction within each site is the mean of from one to three VS values.

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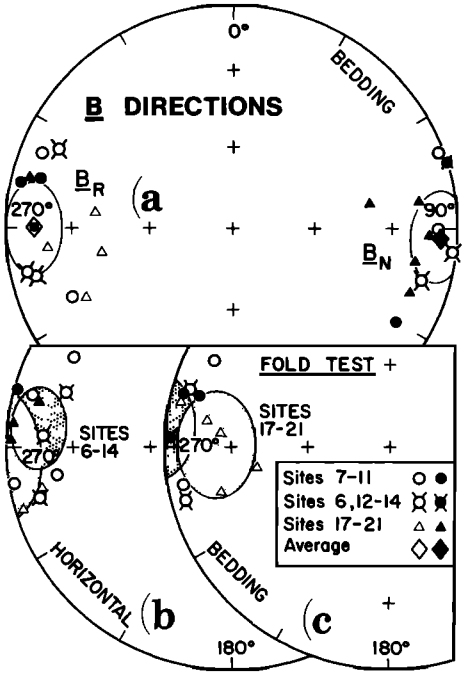


Fig. 13. B_R and B_N site directions (*a*) combined (B_R and B_N meaned together in each site) to illustrate a fold test: (*b*) with respect to horizontal and (*c*) with respect to bedding. The ellipses are the α_{95} errors about the mean directions.

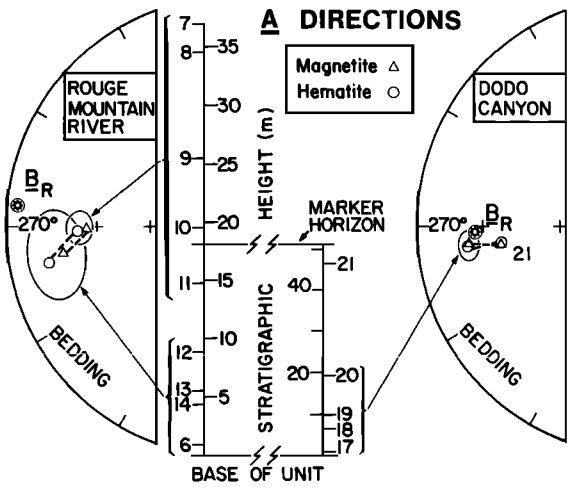


Fig. 14. The change in *A* direction with stratigraphic height in the sequence, and its relation to the B_R direction in the two localities. Note the similar trend shown by both magnetite and hematite. The error circles (α_{95}) about the mean *A* directions are shown.

−13° versus 093°, +09°; Table 7). Taken separately, sites from the Rouge Mountain River locality (6–14) yield a direction with lower inclination (275°, −01°) than do those from the Dodo Canyon locality (17–21) (271°, −23°).

DISCUSSION

The NRM grouping after unfolding is significant at the 95% level of the fold test (Table 8) indicating that the bulk of the magnetization is pre-folding. After cleaning, directions *A* and *C* are found to be prefolding at the 99% level of the fold test (Table 8), but the status of the *B* directions is equivocal, a direction with respect to horizontal being slightly favored.

A is a very stable magnetization, apparently being composed of magnetite (A_M) and hematite (A_H) components. It is probable that the *A* hematite (titanohematite?) is due to oxidation of the *A* magnetite (titanomagnetite?), and that both magnetic components were acquired at relatively the same time.

A major question concerns the relationship between the *A* and *B* hematite magnetizations. The directions do not prove to be significantly different, even though the *B* error circle (α_{95}) does not overlap that of A_H (the error circles of A_H and B_R do overlap slightly). There is in fact excellent agreement between B_R and A_H for the Dodo Canyon limb (~5° apart). In addition, the average intensities of the hematite components are closely similar (A_H, 0.73 ± 0.27; B_R, 0.77 ± 0.49; B_N, 0.68 ± 0.42 mA/m). However, it is my opinion based on several

TABLE 7. Summary of A, B, and C Vectors and Poles With Respect to Bedding

				Directions				Poles				
	Location	*N*	*M*	*D*°, *I*°	*R*	*k*	α_{95}°	°S, °E	*R*	*K*	A_{95}°	δ_p°, δ_m°
A_M	Rouge	9	1.25	265.6, −30.3	8.89	74	6.0	16.7, 139.6	8.93	118	4.8	
	Dodo	5	1.39	263.3, −26.5	4.97	127	6.8	15.6, 142.8	4.98	194	5.5	
	Both	14	1.30	264.8, −29.0	13.85	87	4.3	16.3, 140.8	13.91	139	3.4	2.6, 4.7
A_H	Rouge	9	0.76	263.7, −26.3	8.79	38	8.5	15.5, 142.5	8.87	63	6.5	
	Dodo	5	0.68	263.2, −26.2	4.97	126	6.8	15.5, 142.9	4.99	295	4.5	
	Both	14	0.73	263.5, −26.3	13.76	53	5.5	15.5, 142.6	13.86	92	4.2	3.2, 5.9
A	Rouge	9	···	265.2, −28.4	8.86	59	6.8	15.9, 140.5	8.92	105	5.1	
	Dodo	5	···	263.5, −26.3	4.98	161	6.1	15.5, 142.7	4.99	288	4.5	
	Both	14	···	264.6, −27.7	13.84	79	4.5	15.7, 141.3	13.91	141	3.4	2.7, 4.9
B_R	Rouge	8	0.74	273.8, −05.3	7.47	13	15.8	00.9, 138.6	7.61	18	13.5	
	Dodo	5	0.82	266.4, −24.9	4.65	11	23.6	14.1, 140.4	4.79	19	17.8	
	Both	13	0.77	271.1, −12.8	11.93	11	12.9	06.0, 139.3	12.32	18	10.1	6.7, 13.2
B_N	Rouge	6	0.67	092.6, −01.7	5.63	14	18.8	−02.0, 140.9	5.70	17	17.0	
	Dodo	5	0.68	094.1, +21.0	4.86	28	14.6	08.5, 134.6	4.89	36	12.9	
	Both	11	0.68	093.3, +08.8	10.29	14	12.6	02.9, 138.0	10.53	21	10.2	6.4, 12.7
B	Rouge	8	0.68	274.6, −00.8	7.67	21	12.4	−01.6, 138.7	7.73	26	11.0	
	Dodo	5	0.75	270.5, −22.9	4.84	24	15.8	11.2, 137.4	4.89	35	13.0	
	Both	13	0.71	273.1, −09.3	12.28	17	10.5	03.4, 138.2	12.54	26	8.2	5.3, 10.6
C	Both	12	0.99	291.3, +74.6	11.85	73	5.1	−60.2, 170.1	11.55	25	8.9	8.5, 9.3

A_{95} is the half-angle of cone of confidence about the pole; *k* or *K*, the precision; and δp, δm, the semi-axes of the ellipse of confidence at $P = 0.05$ about the pole calculated from the average directions. See Tables 1 and 2 for explanation of other symbols. Values of *M* are from VS; that under *C* is from 7 determinations from among 33 specimens. A_M, B_R, B_N were obtained from VS; A_H and *C* from vector diagrams.

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TABLE 8. Application of the Fold Test Before (HOR) and After (BDG) Correction for Bedding

	N	Average Precisions (k) HOR (k_1)	BDG (k_2)	k_2/k_1	99% Confidence Limit†	Significance of Test
NRM	14	7.16	16.04	2.24	1.93‡	YES
C	12	14.31	72.55	5.07	4.16	YES
B	13	19.72	16.62	0.84	2.66	NO
A_M	14	13.70	87.13	6.36	2.55	YES
A_H	14	11.87	55.93	4.71	2.55	YES
A	14	12.99	76.89	5.92	2.55	YES

N is sites.
†Values from *McElhinny* [1964].
‡95% confidence limit.

grounds that despite this evidence A_H and B are indeed separate magnetizations. First, the magnetic behavior is indicative of two distinct phases in the T_{UB} range above 600°C (Figure 7*a*). Second, the directions of A_H and B_R for the Rouge Mountain River limb are approximately 20° apart (Figure 14), and there is good overall correspondence between B_R and B_N when one is reversed–much better than that between B_N and A_H. Third, it is natural to expect two hematite phases to be present; namely, (1) hematite alteration of original magnetite detrital particles and (2) red hematite pigment. And finally, evidence indicates that there are three rather than two hematite components remaining after the elimination of A_M.

This latter point was tested by alternately assuming three components to be present (i.e., A_H, B_N, and B_R) and then two (i.e., A_H and B_N). In the former case all three components were determined by VS and their vector sum subtracted from A_M, the AE vector between 500 and 600°C (Method 2, Table 3). In the latter case B_R and A_H were assumed to be equivalent with the better A_H data being used, and B_N was assumed to be the reverse of A_H (Method 3, Table 3). Although both methods gave approximately the same directions (<6° from those of Method 1), Method 2 gave an average intensity (1.27 ± 0.48 mA/m) in much better agreement with that of Method 1 (1.28 ± 0.40 mA/m) than did Method 3 (1.98 ± 0.74 mA/m).

The difference between the directions of A_M determined by Methods 1 and 2 may be largely owing to the paucity of data from which B_R and B_N were determined and to the greater opportunity for introducing errors. It may to a lesser extent be due to inaccuracies in the determination of A_H from VS owing to some overlap of the A_H and B, T_{UB} spectra.

A related model which would explain the presence of A and B components was presented in *Roy and Park* [1974]. According to their model the initial magnetization would be analogously represented by detrital magnetite (A_M), which would be subsequently oxidized to hematite over an extended period thus acquiring the A_H magnetization. Later and (or) concurrent with the acquisition of A_H, hematite pigment would be deposited over a period spanning one or more field reversals and acquire the B_R and B_N magnetizations.

Accepting A and B to be distinct, how is one to explain the varying agreement between A_H and B_R directions at the two localities (Figure 14)? It could be explained by different acquisition times of the pigment in the two areas, which is reasonable since the localities are separated by about 40 km. Alternately, it could be explained by a varying contamination of B in A_H and vice versa. The amount of contamination of at least B in A_H would appear to be small, since A_H for both limbs is <4° from A_M. In addition, A_H does not lie between B_R and A_M as would be expected if biased by B_R (see Figure 14), and always lies on the same side of A_M. This latter fact probably relates more to the sequence in which the major part of each magnetization was acquired rather than to contamination by B. Much of the A_H magnetization may have been acquired through deuteric alteration of magnetite in a pre-existing igneous body, and thus be statistically parallel to the A_M magnetization. The bias between A_H and A_M appears to be superimposed on a consistent trend in the magnetization which is apparently related to stratigraphic height within the sequence in both limbs of the fold (Figure 14), an effect probably due to variation of the field during the deposition of the magnetite particles.

The poles of the A and B magnetizations are plotted in Figure 15. The poles of this study fall close to the Hadrynian Po-

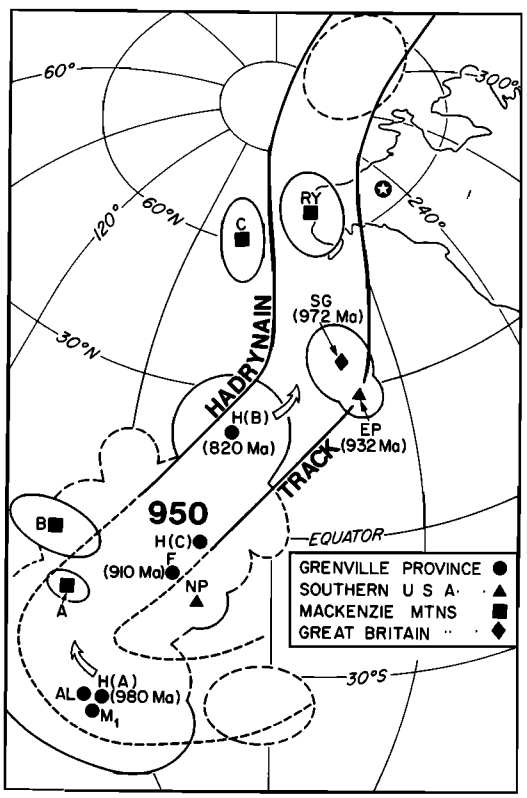


Fig. 15. The A, B, and C poles compared to the Hadrynian Track of *Morris and Roy* [1977]. The lower end of the Hadrynian Track is mainly defined from poles of the Grenville Structural Province of the Canadian Shield. Only what are considered to be key poles are plotted, with the others being represented by the cloud of error circles. The age 950 (Ma) is the calibration mark placed on the track by *Roy and Robertson* [1978] and is predominantly based on the average age of the K-Ar determinations of Grenville rocks. The other ages in parenthesis are considered to be the best age determinations from individual rock units: 820 and 980 Ma from $^{40}Ar/^{39}Ar$ work [*Berger et al.*, 1979], 910 Ma from $^{40}Ar/^{39}Ar$ [*Reynolds*, 1978], 972 Ma from Rb-Sr [*Moorbath*, 1969], and 932 Ma [quoted in *Spall*, 1971]. Poles depicted are: AL, Allard Lake anorthosite [*Hargraves and Burt*, 1967]; EP, rocks of El Paso, Texas [*Spall*, 1971]; F, Frontenac axis dikes [*Park and Irving*, 1972]; H(A), H(B), H(C), the A, B, and C components of the Haliburton intrusions [*Buchan and Dunlop*, 1976]; M_1, Morin complex [*Irving et al.*, 1974]; NP, Nankoweap Formation [*Elston and Grommé*, 1974]; RY, Y component of Rapitan Group [*Morris*, 1977]; SG, Stoer Group [*Stewart and Irving*, 1974]. Error ellipses about A, B, and C have semi-axes p, m as defined in Table 7. Asterisk is the sampling area.

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lar Track proposed by *Morris and Roy* [1977] and illustrate the same trend if–as proposed–*A* predates *B*. Since most of the poles which define this track are secondary and hence undated, only those poles thought to be reasonably well dated have been generally plotted. Poles not plotted are represented by error ellipses. The cloud of error ellipses near the equator represent secondary magnetizations probably acquired during uplift of the Grenville Structural Province following deep burial, and are roughly dated by the K/Ar ages which average about 950 Ma. *Roy and Robertson* [1978] use this age as a calibration mark for the Hadrynian Track at the equator (Figure 15). A more precise method for dating magnetizations involving $^{40}Ar/^{39}Ar$ is now available, which is especially suitable for use where slow cooling following uplift and erosion has occurred. Poles H(A) and H(B) from the Haliburton intrusions [*Buchan and Dunlop,* 1976] and F from the Frontenac axis dikes [*Park and Irving,* 1972] have been dated by this method using the same samples for both radiometric and paleomagnetic work. F, about 20° east of *A*, is dated at ca. 910 Ma [*Reynolds,* 1978]; its age is reasonably consistent with an interpolated value between H(A) (980 Ma) and H(B) (820 Ma) [*Berger et al.,* 1979]. Northeast of these poles are the poles of the Stoer Group (SG) (rotated for closure of the North Atlantic) and the rocks of El Paso, Texas (EP) which have been dated by the Rb-Sr method at 972 ± 23 Ma ($\lambda = 1.42 \times 10^{-11}$ yr^{-1}) [*Moorbath,* 1969] and 932 ± 13 Ma respectively. They appear to have ages inconsistent with the $^{40}Ar/^{39}Ar$ dated poles. There is no reason to fault these Rb-Sr dates, but many do question Rb-Sr dates obtained from sedimentary rocks since they can be ambiguous [*Perry and Turekian,* 1974; *Chaudhuri,* 1976]. Using a simple interpolation between the consistent $^{40}Ar/^{39}Ar$ dated poles, the *A* magnetization would date between about 900 and 950 Ma. This age range fits centrally between the rather broad age limits (1100–800 Ma) placed on the Little Dal fossils [*Hofmann and Aitken,* 1979].

The pole of the *C* direction may represent a magnetization acquired at any time between ca. 950 and 55 Ma. There is good evidence that *C* is carried by goethite, perhaps as a secondary component due to weathering. The magnetization is probably constrained to the late Cretaceous, since any goethite formed prior to early Cambrian would presumably have been dehydrated during deep burial. Goethite would have had opportunity to form and be magnetized in the Cretaceous during uplift prior to folding when these rocks were again accessible to ground water. The pole lies close to the RY pole of *Morris* [1977] from the later Rapitan Group of the same area. Both poles are near the Cretaceous pole for North America (64°N,187°E). It appears unlikely then that the *C* pole is associated with the Hadrynian track of *Morris and Roy* [1977].

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